

# Public whole-genome demonstration

## HG003 · Genome in a Bottle

Sanitized public demonstration generated from an end-to-end GenomeLens analysis of publicly released Genome in a Bottle data.  
This public report contains a selected allowlist of results from the analysis. It is not the production GenomeLens report and does not expose the private analysis engine.

### DATASET PROVENANCE

|                                     |                                  |                                    |                                   |
|-------------------------------------|----------------------------------|------------------------------------|-----------------------------------|
| SOURCE<br><b>Genome in a Bottle</b> | REFERENCE BUILD<br><b>GRCh38</b> | BENCHMARK RELEASE<br><b>v4.2.1</b> | BENCHMARK SCOPE<br><b>chr1-22</b> |
|-------------------------------------|----------------------------------|------------------------------------|-----------------------------------|

### EXTERNAL BENCHMARK

|                              |                                    |                        |                                    |
|------------------------------|------------------------------------|------------------------|------------------------------------|
| CONCORDANT<br><b>187/187</b> | EXPLICIT TRUTH CALLS<br><b>187</b> | MISMATCHES<br><b>0</b> | NORMALIZATION FAILURES<br><b>0</b> |
|------------------------------|------------------------------------|------------------------|------------------------------------|

Concordance is measured on positions where the truth set states a genotype, inside the defined confident regions. It is not whole-genome accuracy, not clinical validation, and not an endorsement by NIST or GIAB.

### SELECTED GENOMIC RESULTS

| FINDING                                                                                    | CATEGORY         | DISPOSITION | EVIDENCE BASIS                               |
|--------------------------------------------------------------------------------------------|------------------|-------------|----------------------------------------------|
| Celiac Disease Susceptibility (second-most-common allele) carrier -- susceptibility_allele | carrier          | unresolved  | curated clinical evidence                    |
| Below-median longevity composite (improvement headroom)                                    | longevity        | conditional | published association evidence               |
| CYP2C19 Rapid Metabolizer (RM) (clopidogrel)                                               | pharmacogenomics | conditional | published pharmacogenomic guideline evidence |

A representative selection, one per category. The full sanitized analysis contains more results than are shown here.

HEALTH-ECONOMIC RESULTS

|                                            |                                                |                                                                                                       |                                    |
|--------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------|
| REFERENCE-CASE<br>NMB<br><b>\$1,497.24</b> | CANONICAL<br>EXPECTED NMB<br><b>\$3,588.00</b> | INCREMENTAL COST<br><b>\$388.53</b>                                                                   | INCREMENTAL QALYS<br><b>0.0189</b> |
| ICER<br><b>20,603</b>                      | PROBABILITY COST-EFFECTIVE<br><b>80.7%</b>     | NOTE<br>Reference-case and canonical pathway results answer different questions and are never summed. |                                    |

**Evidence basis.** Genomic findings rest on curated clinical evidence; published association evidence; published pharmacogenomic guideline evidence. The economic figures are modeled estimates built on published evidence and stated assumptions, not realized savings.

**For nontechnical readers.** These results are a research demonstration of what GenomeLens can produce from real whole-genome data. The benchmark measures agreement only at the specific GIAB locations tested and is not a claim of whole-genome or clinical accuracy. The health-economic figures are modeled estimates based on published evidence and assumptions; they are not a diagnosis, medical advice, or a statement that a genome is literally “worth” a particular dollar amount.

SCOPE AND LIMITATIONS

- Public demonstration output; not clinical validation.
- Benchmark scope is limited to the stated GIAB panel and chr1-22.
- Modeled economics are not realized savings.
- Representative findings only; the full sanitized report is the complete output.